United International University
QUEST FOR EXCELLENCE

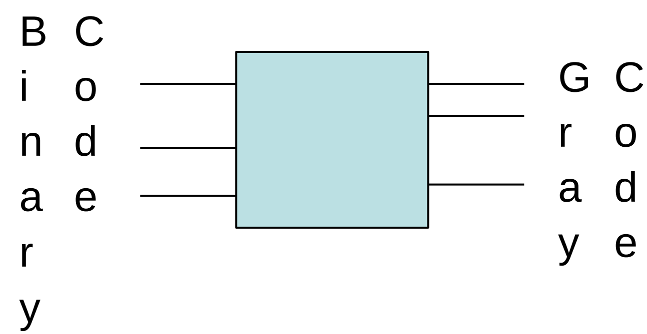
CSE 1326: Digital Logic Design Lab
3-Bit Binary to Gray code

United International University

What to do

1. **Problem Description**: Find out the relation between 3-bit binary to gray code. The circuit will take a binary code as input, and provide the corresponding gray code as output.
2. Derive the truth table from problem description.

	Binary A B C	Gray Code X Y Z
m0	**0 0 0**	**0 0 0**
m1	**0 0 1**	**0 0 1**
m2	**0 1 0**	**0 1 1**
m3	**0 1 1**	**0 1 0**
m4	**1 0 0**	**1 1 0**
m5	**1 0 1**	**1 1 1**
m6	**1 1 0**	**1 0 1**
m7	**1 1 1**	**1 0 0**



How to proceed

3. Now, derive the functions from Truth Table (TT)

- X = F (A, B, C) = ??? = sum of minterms
- Y = F (A, B, C) = ??? = sum of minterms
- Z = F (A, B, C) = ??? = sum of minterms

4. Minimize/Optimize/Simplify functions: (Boolean algebra or Karnaugh Maps)
    - Draw K-map according to # of minterms
    - Populate 1's and 0's
    - Group 1's in the number of power of 2 using rectangles. The larger the rectangles, the better
    - Write function: You will get a term from each rectangle.

How to proceed

4. Build the circuit in LogiSim and Trainer board.
5. Verify the circuit according to the truth table.

Writing Report

- The report should contain the following:
  - Decimal and Binary representation of 3-Bit Gray code
  - The K-maps you have drawn and the minimized functions for X, Y and Z
  - The circuit diagram
  - List of ICs you used